CHERN CLASSES OF THE DUAL OF A FINITE LOCALLY FREE MODULE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as three spans, lines 8-12 (the verbatim package together with the project's own reference, slogan and history display-environment definitions, including the author's commented-out older form of the reference environment), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). The lines 8-12 span is required because a counted statement or its surrounding text may use those display environments, which the project typesets as comments. Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Editorial note. The author's closing sentence at source lines 8562-8563, 'The result follows from a formal computation which we omit.', is retained verbatim at the end of the counted proof. The reduction by the splitting principle, the construction of the dual filtration and the two product expressions are all inside the span; only the expansion of the products is left to the reader.

Allowed standard prerequisites (editorial).

- The splitting principle: a morphism over which the module acquires a filtration by invertible modules, with injective pullback on Chern classes. Same chapter chow.tex, label lemma-splitting-principle, invoked at proof line 8532. Produces the morphism to which one may pull back, since the pullback on the relevant groups is injective after any base change, which is what allows the relation to be checked after a filtration exists.
- The total Chern class of a module filtered by invertible modules as the product of the linear factors, and additivity in the invertible sheaf. Same chapter chow.tex, labels lemma-chern-filter-by-linebundles (proof line 8555) and lemma-c1-cap-additive (proof line 8554). The first writes the total Chern classes of the module and of its dual as products of linear factors in the first Chern classes of the successive quotients and of their inverses, and the second gives that the first Chern class of an inverse is the negative of the first Chern class.

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Source reference. Original lemma 02SP: sources/stacks-chow/pinned-original-source.tex, source lines 4330–4330. Navigation pointer for \ref{lemma-c1-cap-additive} used inside the counted span (line 8554); the argument that consumes it is retained in the span.

Source reference. Original lemma 02UJ: sources/stacks-chow/pinned-original-source.tex, source lines 8022–8022. Navigation pointer for \ref{lemma-chern-filter-by-linebundles} used inside the counted span (line 8555); the argument that consumes it is retained in the span.

Source reference. Original lemma 02UL: sources/stacks-chow/pinned-original-source.tex, source lines 8429–8429. Navigation pointer for \ref{lemma-splitting-principle} used inside the counted span (line 8532); the argument that consumes it is retained in the span.

Source reference. Original reference 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 8521); the argument that consumes it is retained in the span.

Lemma 0.1. *In Situation 02QL let X be locally of finite type over S. Let $\mathcal{E}$ be a finite locally free $\mathcal{O}_X$-module with dual $\mathcal{E}^\vee$. Then*

$$c_i(\mathcal{E}^\vee) = (-1)^i c_i(\mathcal{E})$$

in $A^i(X)$.

Proof. Choose a morphism $\pi : P \to X$ as in Lemma 02UL. By the injectivity of π^* (after any base change) it suffices to prove the relation between the Chern classes of $\mathcal{E}$ and $\mathcal{E}^\vee$ after pulling back to P. Thus we may assume there exist invertible $\mathcal{O}_X$-modules $\mathcal{L}_i$, $i = 1, \ldots, r$ and a filtration

$$0 = \mathcal{E}_0 \subset \mathcal{E}_1 \subset \mathcal{E}_2 \subset \ldots \subset \mathcal{E}_r = \mathcal{E}$$

such that $\mathcal{E}_i/\mathcal{E}_{i-1} \cong \mathcal{L}_i$. Then we obtain the dual filtration

$$0 = \mathcal{E}_r^\perp \subset \mathcal{E}_1^\perp \subset \mathcal{E}_2^\perp \subset \ldots \subset \mathcal{E}_0^\perp = \mathcal{E}^\vee$$

such that $\mathcal{E}_{i-1}^\perp/\mathcal{E}_i^\perp \cong \mathcal{L}_i^{\otimes -1}$. Set $x_i = c_1(\mathcal{L}_i)$. Then $c_1(\mathcal{L}_i^{\otimes -1}) = -x_i$ by Lemma 02SP. By Lemma 02UJ we have

$$c(\mathcal{E}) = \prod\nolimits_{i=1}^r (1 + x_i) \quad \text{and} \quad c(\mathcal{E}^\vee) = \prod\nolimits_{i=1}^r (1 - x_i)$$

in $A^*(X)$. The result follows from a formal computation which we omit. □